

MODULE 4: PRODUCTION PLANNING AND CONTROL (PPC)

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Production Planning and Control (PPC)

Production Planning and Control (PPC) is a critical function in industrial management that involves the planning, coordination, and supervision of the production process. It ensures that manufacturing operations run smoothly, efficiently, and effectively to meet customer demands while optimizing resource utilization. **Key Components of PPC:**

1. **Production Planning:** Involves forecasting demand, determining production capacity, and creating detailed production schedules. It includes:
 - **Forecasting:** Estimating future demand for products.
 - **Capacity Planning:** Assessing the production capacity needed to meet demand.
 - **Routing:** Establishing the most efficient path for production processes.
 - **Scheduling:** Allocating resources and setting timelines for production tasks.
2. **Production Control:** Focuses on monitoring and adjusting the production process to ensure it aligns with the production plan. It includes:
 - **Dispatching:** Issuing orders to begin production.
 - **Monitoring:** Tracking progress and performance metrics to ensure adherence to the schedule.
 - **Corrective Actions:** Adjusting resolve issues and ensure production targets are met.
 - **Feedback Loop:** Collecting data from production processes to inform future planning and decision-making.

Techniques Used in PPC

1. **Material Requirement Planning (MRP):** A system that calculates material requirements based on production schedules, ensuring that the right materials are available at the right time.
2. **Just-In-Time (JIT):** A strategy that aims to minimize inventory costs by receiving goods only as they are needed in the production process.
3. **Lean Manufacturing:** A methodology focused on eliminating waste and improving efficiency in production processes.
4. **Critical Path Method (CPM):** A project management technique that identifies the longest sequence of dependent tasks, allowing for effective scheduling and resource allocation.

5. **Gantt Charts:** Visual tools used to represent production schedules, helping managers track progress and deadlines.

Need and Importance of PPC

1. **Efficient Resource Utilization:** PPC ensures that resources such as labor, materials, and machinery are used effectively, minimizing waste, and maximizing productivity.
2. **Meeting Customer Demand:** By accurately forecasting demand and scheduling production, PPC helps organizations meet customer needs and avoid stockouts or overproduction.
3. **Quality Management:** Effective PPC helps maintain quality standards by monitoring production processes and ensuring adherence to specifications, thereby reducing defects.
4. **Cost Control:** PPC aids in identifying cost-saving opportunities and managing production expenses, contributing to overall profitability.
5. **Flexibility and Adaptability:** A well-structured PPC system allows organizations to respond quickly to changes in market demand or supply chain disruptions, maintaining competitive advantage.
6. **Improved Coordination:** PPC facilitates better communication and coordination between different departments, ensuring alignment in production goals and schedules.

Functions of PPC

1. **Forecasting:** Estimating future demand for products based on historical data, market trends, and customer input to guide production planning.
2. **Capacity Planning:** Assessing the production capacity needed to meet forecasted demand, ensuring that the organization has the necessary resources in place.
3. **Scheduling:** Developing detailed production schedules that outline when and how much of each product will be produced, including labor and equipment allocations.
4. **Routing:** Determining the optimal path for materials and processes through the production system, ensuring efficiency and minimizing bottlenecks.
5. **Dispatching:** Issuing orders to start production and allocating resources as needed to ensure that production plans are executed effectively.
6. **Monitoring and Control:** Continuously tracking production processes to ensure they align with plans and making necessary adjustments to address deviations or issues.
7. **Quality Control:** Implementing quality assurance measures throughout the production process to maintain product standards and minimize defects.
8. **Feedback and Adjustment:** Collecting performance data to evaluate production outcomes, enabling continuous improvement in planning and control processes.

Challenges in PPC

1. **Demand Variability:** Fluctuations in customer demand can complicate planning and require quick adjustments to production schedules.
2. **Resource Constraints:** Limited availability of materials, labour, or machinery can hinder production and affect timelines.
3. **Quality Control:** Maintaining consistent product quality can be challenging, especially in large-scale production.
4. **Supply Chain Disruptions:** Unexpected interruptions in the supply chain can lead to delays and affect the overall production process.
5. **Technological Changes:** Rapid advancements in technology can require constant adaptation of production processes and equipment.

Types of Production

1. **Job Production:** Production of custom products tailored to specific customer requirements. **Example:** A tailor making a bespoke suit or a specialized machine manufactured for a particular client. General Approach:
 - **Customization:** Focus on specific customer requirements and tailor the production process accordingly.
 - **Flexibility:** Use flexible manufacturing systems that can quickly adapt to different job specifications.
 - **Small Scale:** Typically involves small production runs; prioritize high-quality craftsmanship.
 - **Monitoring:** Closely monitor each job's progress to ensure adherence to quality and specifications.
2. **Batch Production:** Production of goods in groups or batches, allowing for variations in the product. **Example:** A bakery producing a batch of bread or cookies, where different flavours or styles are created in distinct batches. General Approach:
 - **Group Production:** Produce goods in batches, allowing for some variety in product types.
 - **Scheduling:** Schedule production runs based on demand forecasts, considering setup times for different batches.
 - **Inventory Management:** Manage inventory levels to ensure the right materials are available for each batch.
 - **Quality Checks:** Implement quality control measures after each batch to ensure consistency.
3. **Mass Production:** High-volume production of standardized products, often using assembly line techniques. **Example:** Automobile manufacturing, where thousands of identical vehicles are produced using automated machinery. General Approach:
 - **Standardization:** Focus on producing large quantities of standardized products using assembly line techniques.
 - **Automation:** Utilize automated machinery to improve efficiency and reduce production costs.
 - **Continuous Monitoring:** Regularly monitor production metrics to maintain output levels and quality.
 - **Lean Techniques:** Apply lean manufacturing principles to minimize waste and optimize workflows.
4. **Continuous Production:** Production process that runs continuously to produce highly standardized products. **Example:** Oil refining or the production of chemicals, where the process operates 24/7 to maintain efficiency. General Approach:
 - **Uninterrupted Flow:** Maintain a continuous flow of production with minimal downtime.
 - **Process Automation:** Implement advanced automation and control systems to monitor and manage processes.
 - **Maintenance Scheduling:** Plan regular maintenance to prevent unexpected disruptions.
 - **Data Analysis:** Use real-time data analysis to optimize processes and respond to any deviations promptly.

5. **Project Production:** Production undertaken for a specific project or outcome, often with unique characteristics. **Example:** Construction of a skyscraper or a large infrastructure project like a bridge or dam. General Approach:
- **Planning:** Develop detailed project plans that outline the scope, timelines, and resource requirements.
 - **Coordination:** Ensure coordination among various stakeholders involved in the project.
 - **Milestones:** Set clear milestones and deliverables to track progress.
 - **Risk Management:** Identify potential risks and develop contingency plans to mitigate them.

Forms Used in Production Planning and Control (PPC)

1. **Production Schedule:** Specifies production timelines for efficient resource use and timely delivery. Approach: Use Gantt charts to plan start and finish dates.
2. **Material Requirement Planning (MRP):** Lists needed materials to avoid delays. Approach: Use forecasts and schedules to calculate material needs, considering lead times.
3. **Work Orders:** Detailed instructions for production tasks, including materials, equipment, and time. Approach: Create comprehensive work orders specifying requirements.
4. **Capacity Planning Reports:** Ensures production capacity meets demand to avoid bottlenecks. Approach: Compare capacity with demand forecasts and adjust resources.
5. **Inventory Management:** Tracks inventory levels for optimal stock. Approach: Perform audits and use tracking software for accuracy.
6. **Quality Control Checklists:** Ensures products meet standards and reduces defects. Approach: Create checklists and conduct inspections throughout production.
7. **Performance Reports:** Measures production efficiency for improvement. Approach: Collect and analyse KPIs like cycle time and defect rates.

Scheduling

Scheduling refers to the process of assigning specific time slots for tasks or activities within a production system. It involves determining when each task should start and finish, the resources required, and the sequence of operations. Effective scheduling ensures that production processes run smoothly, efficiently, and on time.

Need for Scheduling:

1. **Enhances Productivity:** Maximizes output by ensuring resources are used effectively, leading to efficient operations.
2. **Reduces Idle Time:** Minimizes periods when workers or machinery are not engaged, promoting continuous production flow.
3. **Improves Delivery Timeliness:** Ensures products are completed on schedule, enhancing customer satisfaction and meeting commitments.
4. **Facilitates Resource Management:** Allocates materials and labour efficiently, reducing bottlenecks and ensuring availability when needed.
5. **Enables Flexibility:** Allows quick adjustments to changes in demand or production delays, maintaining operational continuity.
6. **Supports Strategic Planning:** Provides data for analysing performance, aiding informed decisions on future production strategies and capacity.

Gantt Chart

A Gantt chart is a visual scheduling tool that represents project tasks along a timeline. It provides a clear overview of task durations, start and end dates, and the sequence of activities, making it easier to monitor progress and manage resources. **Format of a Gantt Chart:**

- **Horizontal Axis:** Represents time, usually displayed in days, weeks, or months.
- **Vertical Axis:** Lists tasks or activities to be completed.
- **Bars:** Horizontal bars represent the duration of each task, indicating when it starts and ends.
- **Milestones:** Symbols or markers may indicate significant points in the project timeline (e.g., project start, completion dates).
- **Dependencies:** Arrows or lines may connect tasks to show relationships or dependencies between them.

Method to Prepare a Gantt Chart:

1. **Define the Project Scope:** Identify the objectives, deliverables, and major tasks required to complete the project.
2. **List All Tasks:** Break down the project into smaller tasks or activities and list them in the order they need to be completed.
3. **Estimate Task Durations:** Determine how long each task will take to complete. This can be based on historical data, expert judgment, or similar projects.
4. **Identify Dependencies:** Analyse the relationships between tasks to determine which tasks must be completed before others can begin.
5. **Create the Chart:**
 - Draw a horizontal timeline along the top of a chart and list the tasks vertically along the left side.
 - For each task, draw a horizontal bar that spans the estimated duration of the task, placing it on the timeline according to its start and end dates.
 - Add milestones and dependencies as needed.
6. **Review and Adjust:** Review the Gantt chart for accuracy and completeness. Adjust task durations or start dates as necessary based on resource availability or changes in project scope.
7. **Monitor Progress:** Regularly update the Gantt chart as tasks are completed, ensuring it reflects the status of the project.

Critical ratio scheduling method

Critical ratio (CR) is an index number computed by dividing the time remaining until due date by the work time remaining.

CR is easy to calculate and performs better than other priority rules (FCFS, SPT, LPT, EDD) when it comes to dealing with job lateness criteria.

$$CR = \frac{TIME\ REMAINING}{WORKDAYS\ REMAINING} = \frac{Due\ Date - Today's\ Date}{Work\ Time\ Remaining}$$

Bottlenecking

Meaning: A bottleneck occurs when a particular stage in a production or operational process limits the overall capacity or speed of the system. It is the point where the demand placed on a resource exceeds its capacity, leading to delays and inefficiencies.

Effects of Bottlenecking:

1. **Reduced Productivity:** Overall output is constrained by the bottleneck, causing slower production rates.
2. **Increased Lead Time:** The time it takes for products or services to move through the process increases.
3. **Higher Costs:** Due to inefficiencies and delays, operational costs can rise, affecting profitability.
4. **Imbalance in Workflows:** Bottlenecks cause an uneven distribution of tasks, where some resources are underutilized, while others are overwhelmed.
5. **Customer Dissatisfaction:** Delays in production or services can lead to unmet deadlines and unhappy customers.
6. **Reduced Flexibility:** Bottlenecks limit the ability to adapt quickly to changes in demand or production schedules.

Ways to Reduce Bottlenecking:

1. **Identify Bottlenecks:** Use techniques like process mapping, flow charts, or real-time monitoring to locate the bottleneck.
2. **Optimize Bottleneck Processes:** Invest in improving the capacity or efficiency of the bottlenecked stage, either through automation or reorganization.
3. **Add More Resources:** Allocate additional resources such as personnel, machines, or time to the bottleneck stage.
4. **Balance Workload Distribution:** Evenly distribute the workload across all stages to prevent overloading a single point.
5. **Reduce Variability:** Minimize variability in input or processes to maintain a steady flow through the bottleneck.
6. **Improve Process Design:** Rethink or redesign the production process to remove or lessen the impact of the bottleneck.
7. **Implement Lean Techniques:** Apply lean principles such as Just-In-Time (JIT) production to optimize workflow and reduce inefficiencies.
8. **Increase Training:** Provide specialized training to workers at bottleneck points to increase speed and efficiency.
9. **Use Buffer Inventory:** Keep buffer stock to reduce the effect of delays caused by bottlenecks.
10. **Continuous Monitoring and Improvement:** Regularly review processes to identify new or evolving bottlenecks and address them proactively.

Critical ratio scheduling method and numeric examples.

Scheduling using Gantt Chart (for at least 5-7 components having 5-6 machining operations, with processes, setting and operation time for each component and process, resources available, quantity and other necessary data),

At least two examples.